

Week 9: Learning – Bayesian Learning

Practical Assignment

For this practical, we revisit the simple regression example of Question 2 in Week 8 (fitting a line to noisy measurements), and we perform Bayesian parameter estimation in the following two ways:

- MAP estimation to determine parameter values that have the highest likelihood in the posterior distribution over the parameters
- The full Bayesian, where we determine the posterior distribution over the parameters using (standard) inference.

1 MAP estimation for a simple regression example

For this question, we revisit the line fitting problem of Question 2 of Week 8, and use MAP estimation to determine the y -coordinates of the two endpoints of the line, y_I and y_F . In this part you do not necessarily have to use `emdw`'s abilities directly (although you can). You can therefore do your coding in whichever environment that suits you (e.g. Python). [To help you check your work, the answer at an intermediate point is given at the end of this question in blue.](#)

- (On paper) Redraw the BN for this model with M measurements using plate notation.
- (On paper) Choose the prior parameter distribution as $p(\Theta) = \mathcal{N}(y_I; \mu_I, \sigma_I^2) \cdot \mathcal{N}(y_F; \mu_F, \sigma_F^2)$. Use the likelihood function $p(\mathcal{D}|\Theta)$ from Week 8 to write $\log p(\Theta|\mathcal{D})$ as a sum of quadratic terms in terms of $\hat{y}^{[m]}$, y_I and y_F (except for a few extra terms, your answer should be the same as for Question 2(b) of Week 8).
- (On paper) Now derive $\Theta^{\text{MAP}} = \arg \max_{\Theta} \log p(\Theta|\mathcal{D})$ using a similar procedure to that of Question 2(c) of Week 8 (you should be able to reuse much of that answer).
- (In code) Use the same procedure as Question 2(d) of Week 8 to generate data (or reuse the same data). Choose appropriate parameters for the prior distribution $p(\Theta)$ and calculate Θ^{MAP} for the generated data. Plot the actual line, data points, MAP estimate as well as ML estimate on the same graph. How similar are the MAP and ML estimates?
- (In code) Repeat the sampling of measurements and subsequent parameter estimation, but vary the number of measurements. What is the smallest number of measurements that you need to calculate a MAP estimate? What is the effect of the number of measurements on the difference between the MAP and ML estimates?
- (In code) Repeat the sampling of measurements and subsequent parameter estimation, but vary the measurement noise variance σ_v^2 while keeping the number of measurements fixed. How does this parameter influence the difference between the MAP and ML estimates?

Answers:

(b)

$$\begin{aligned} \log p(\Theta|\mathcal{D}) = & -\frac{1}{2\sigma_I^2} (y_I - \mu_I)^2 - \frac{1}{2\sigma_F^2} (y_F - \mu_F)^2 \\ & + \sum_{m=1}^M \left[-\frac{1}{2\sigma_v^2} \left(\hat{y}^{[m]} - (1 - \alpha^{[m]})y_I - \alpha^{[m]}y_F \right)^2 \right] \\ & + \text{terms that do not contain } y_I \text{ or } y_F \end{aligned}$$

2 Full Bayesian estimation for a simple regression example

Finally, let's do full Bayesian estimation of the same line fitting example we have used for MLE and MAP estimation. For full Bayesian estimation, we are interested in calculating the posterior *distribution* over the parameters y_I and y_F instead of calculating *values* for them. Specifically, we want to determine the posterior distribution $p(\Theta|\mathcal{D}) = p(y_I, y_F | \hat{y}^{[1]}, \dots, \hat{y}^{[M]})$.

Although we can use `emdw` (you are welcome to try!), it is easier for this simple problem to symbolically derive an expression for the posterior distribution and then implement it in software that can perform matrix operations and visualise data. You can therefore again do your coding in whichever environment that suits you (e.g. Python). [To help you check your work, the answer at an intermediate point is given at the end of this question in blue.](#)

- (On paper) Redraw the BN for this model with M measurements using plate notation, but with y_I and y_F drawn as random variables, not as parameters. Now write the posterior distribution $p(\Theta|\mathcal{D})$ in terms of the factors in this model.
- (On paper) Write each factor in the model as a canonical Gaussian factor in terms of the random variables y_I and y_F . Use the same distributions as in Question 1(b) for the prior distributions over y_I and y_F . For the factors describing the relationship between the measurement $\hat{y}^{[m]}$ and the y -coordinates of the endpoints y_I and y_F , use the measurement model as described in Question 2 of Week 8, and write it as a conditional distribution in the mean-covariance form as we have done multiple times in Week 7. Then convert it into the canonical form in terms of random variables y_I and y_F using similar manipulations as for the example in Week 7's notes (or use the linear Gaussian result on the formula sheet).
- (On paper) Now calculate the posterior distribution $p(\Theta|\mathcal{D})$ by combining the canonical Gaussian factors of (b). The parameters of the posterior distribution should be very similar to the matrices and vectors you calculated for the MAP estimation in Question 1.
- (In code) Calculate the information matrix $\mathbf{K}$ and information vector $\mathbf{h}$ for the posterior distribution $p(\Theta|\mathcal{D})$ using the same data and parameter choices as for Question 1. Convert the posterior distribution to the mean-covariance form by calculating the mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. Does this posterior distribution make sense given the data and prior distributions?
- (In code) Draw a number of lines $\begin{bmatrix} y_I & y_F \end{bmatrix}^T$ from the posterior distribution $p(\Theta|\mathcal{D})$ and plot them with the actual line, data points, as well as the ML and MAP estimates on the same graph.
- (In code) Repeat the sampling of measurements and subsequent parameter estimation, but vary the number of measurements as well as the variance of the measurement noise. For which scenarios does the full Bayesian approach provide significantly more information about the parameters than ML and MAP estimation, and for which scenarios are there no significant differences?
- (In code) Can you retrieve the ML estimate from the posterior distribution $p(\Theta|\mathcal{D})$? Can you retrieve the MAP estimate from the posterior distribution?

Answers:

(b)

$$\begin{aligned}
 p(y_I) &= \mathcal{C}\left(y_I; \frac{1}{\sigma_I^2}, \frac{\mu_I}{\sigma_I^2}, -\right) \\
 p(y_F) &= \mathcal{C}\left(y_F; \frac{1}{\sigma_F^2}, \frac{\mu_F}{\sigma_F^2}, -\right) \\
 p(\hat{y}^{[m]} | y_I, y_F) &= \mathcal{C}\left(\begin{bmatrix} y_I \\ y_F \end{bmatrix}; \frac{1}{\sigma_v^2} \begin{bmatrix} (1 - \alpha^{[m]})^2 & \alpha^{[m]}(1 - \alpha^{[m]}) \\ \alpha^{[m]}(1 - \alpha^{[m]}) & (\alpha^{[m]})^2 \end{bmatrix}, \frac{1}{\sigma_v^2} \begin{bmatrix} (1 - \alpha^{[m]})\hat{y}^{[m]} \\ \alpha^{[m]}\hat{y}^{[m]} \end{bmatrix}, -\right)
 \end{aligned}$$

3 Assessment: Class test

There will be short, written class test at the end of the practical period in the practical venue (see STEMLearn for details). If you have worked through the practical assignment and understood everything, then the test should be easy!